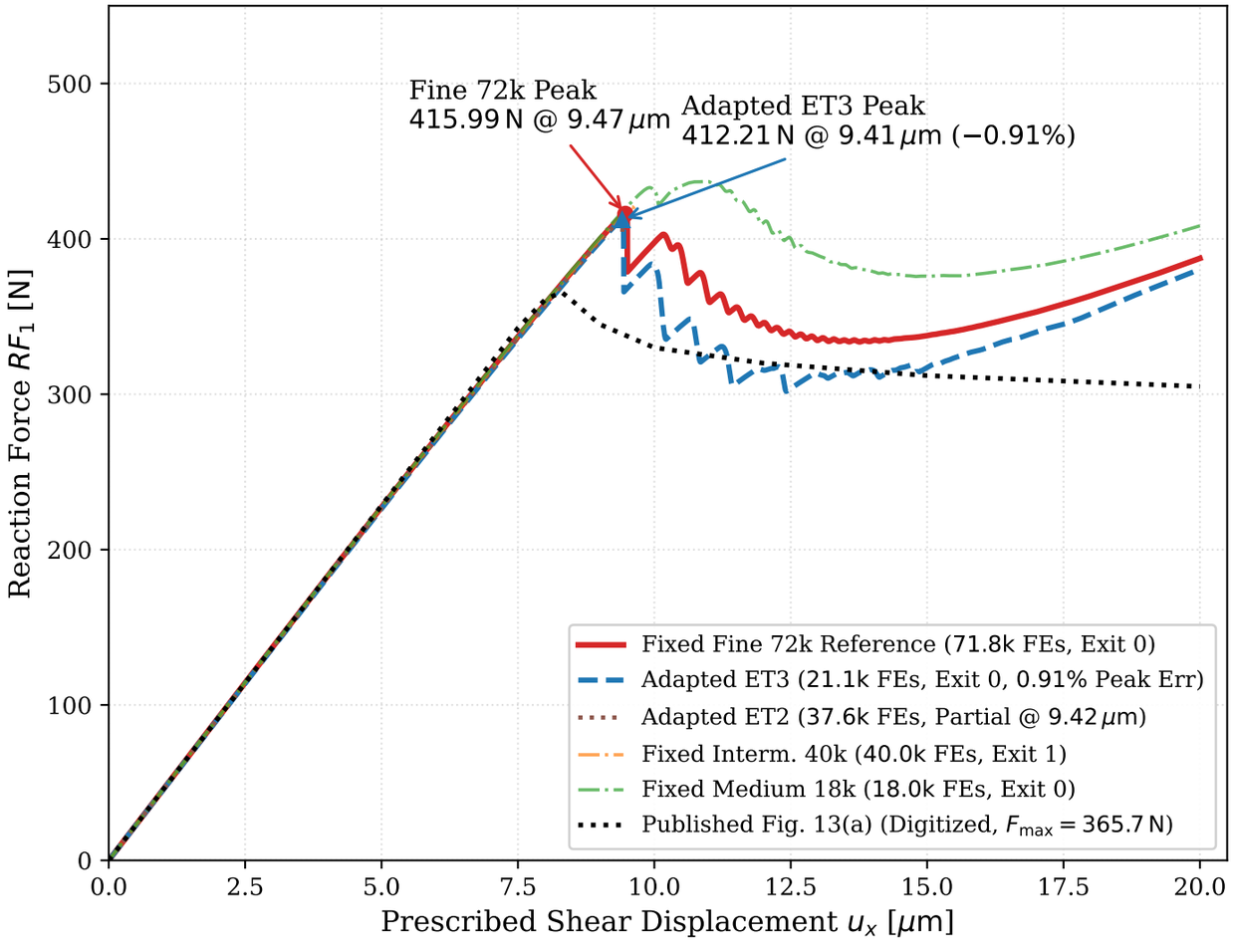
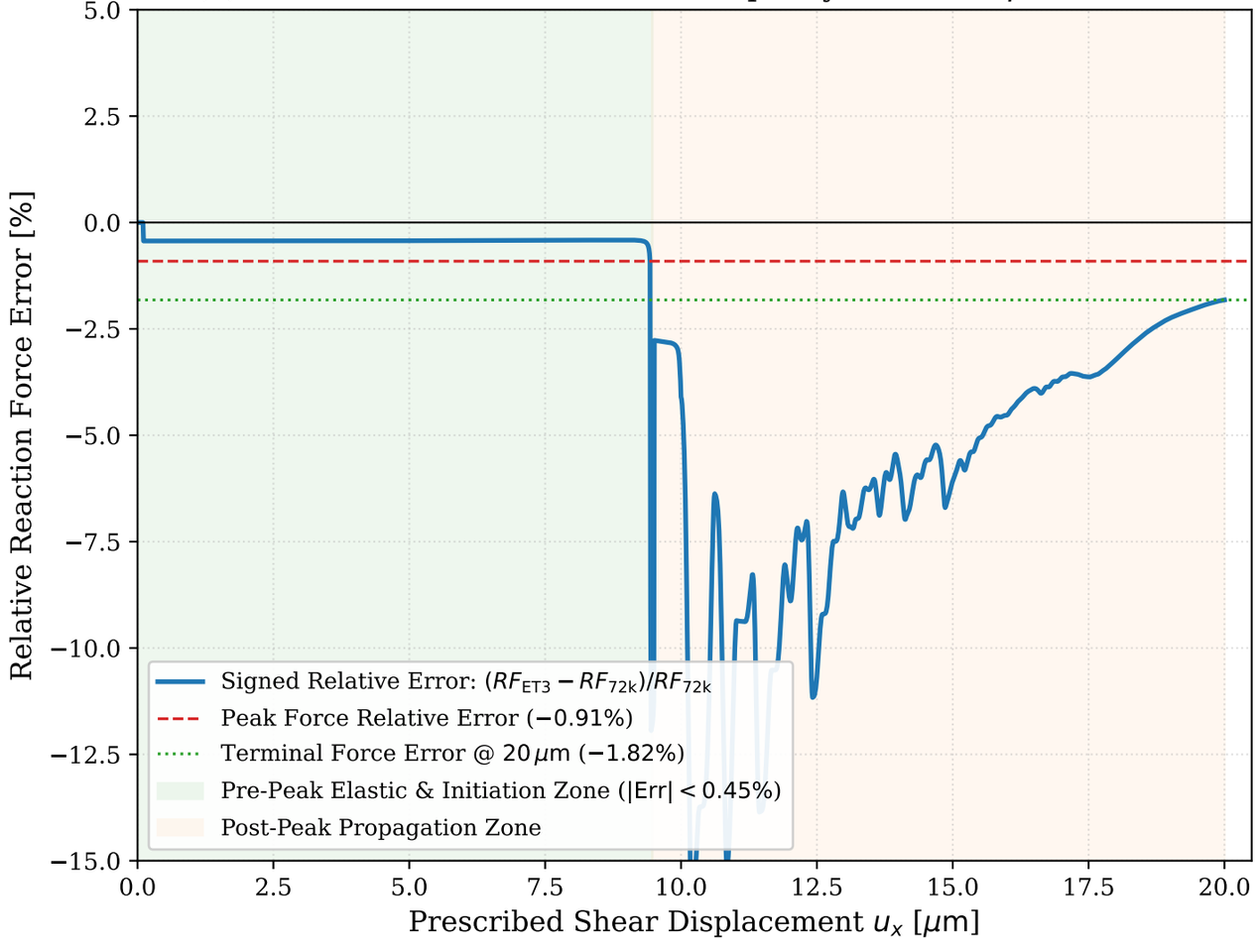
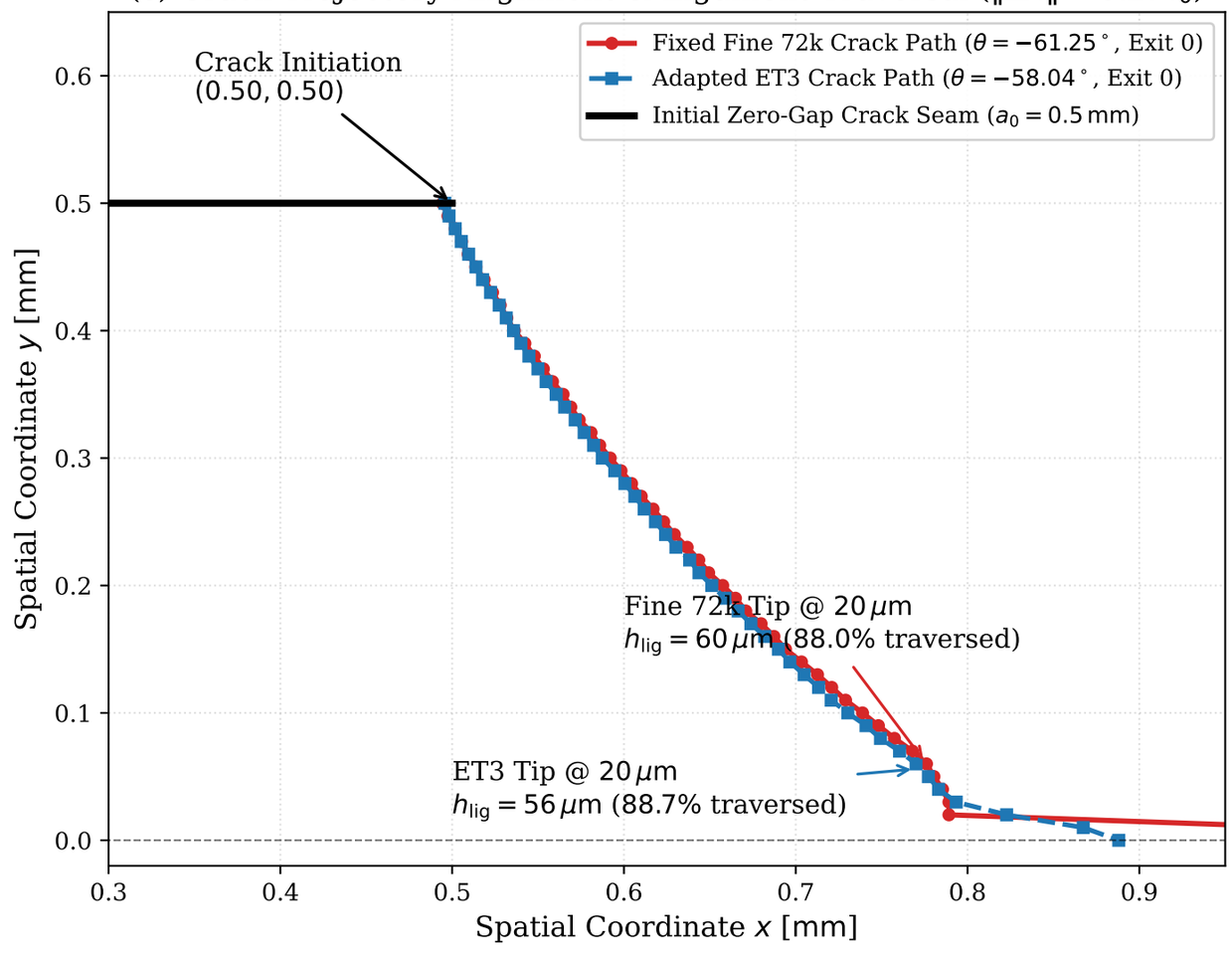
(a) Full Fracture Response: Fixed Fine vs. Adaptive Solutions



(b) Pointwise Relative Force Discrepancy across 20 μm Horizon



(c) Crack Trajectory Alignment & Ligament Traversal ($\|\Delta x\| < 0.27\, l_0$)



(d) Computational Efficiency: 70.7% Elements Reduced, 2.79x Solver Speedu

